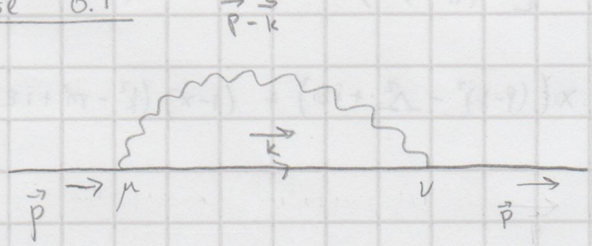
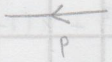
Quantum Field theory Week 6. Homework

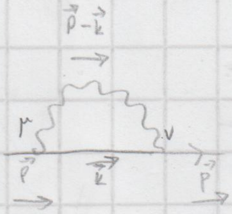
Exercise 6.1



The external legs of this diagram are part of a bigger diagram so we label them as propagato which gives the Feynman rule:

Dirac propagator:  $= \dfrac{i(\not p + m)}{p^2 - m^2 + i\varepsilon}$

In the diagram we have loop of photon. which mean we need to integrate over all the possible momentum k, we get the k-integral $\int \frac{d^4k}{(2\pi)^4}$ integrating the whole thing. The vertex is standard QED-vertex that gives us $-ie\gamma^\mu$. Using the Feynman rules we get



$$= \int \frac{d^4k}{(2\pi)^4} \frac{i(\not p + m)}{p^2 - m^2 + i\varepsilon} (-ie\gamma^\mu) \frac{i(\not k + m)}{k^2 - m^2 + i\varepsilon} (-ie\gamma^\nu) \frac{i(\not p + m)}{p^2 - m^2 + i\varepsilon} \underbrace{\frac{-ig^{\mu\nu}}{(p-k)^2 + i\varepsilon}}_{\text{Photon propagator}}$$

We don't need $i\varepsilon$ prescription for fermion propagator whose momenta we aren't integrating over so we can drop them

The expression for the diagrams is

$$= \frac{i(\not p + m)}{p^2 - m^2} \left[-e^2 \int \frac{d^4k}{(2\pi)^4} \gamma^\mu \frac{(\not k + m)}{k^2 - m^2 + i\varepsilon} \gamma_\mu \frac{1}{(p-k)^2 + i\varepsilon} \right] \frac{i(\not p + m)}{p^2 - m^2}$$

Exercice 6.2

$$-i\Sigma_2(p) \equiv -e^2 \int \frac{d^4k}{(2\pi)^4} \gamma_\mu \frac{\not k + m}{k^2 - m^2 + i\varepsilon} \gamma^\mu \frac{1}{(p-k)^2 + i\varepsilon}$$

Let's regualate Ir divergence by small photon mass μ and UV-divergence by Pauli-Villars:

$$\Rightarrow \frac{1}{(p-k)^2 + i\varepsilon} \rightarrow \frac{1}{(p-k)^2 - \mu^2 + i\varepsilon} - \frac{1}{(p-k)^2 - \Lambda^2 + i\varepsilon}$$

$$\Rightarrow -i\Sigma_2(p) = -e^2 \int \frac{d^4k}{(2\pi)^4} \gamma_\mu (\not k + m) \gamma^\mu \frac{1}{k^2 - m^2 + i\varepsilon} \left[\frac{1}{(p-k)^2 - \mu^2 + i\varepsilon} - \frac{1}{(p-k)^2 - \Lambda^2 + i\varepsilon} \right]$$

Now let's use Feynman parametrization: $\dfrac{1}{AB} = \displaystyle\int_0^1 dx \frac{1}{[xA + (1-x)B]^2}$

$$\Rightarrow \quad -i\Sigma_2(p) = -e^2 \int d^4k/(2\pi)^4 \int_0^1 dx \; \gamma_\mu(\not{k}+m)\gamma^\mu \quad \left[x\left((p-k)^2-\mu^2+i\varepsilon\right)+(1-x)(k^2-m^2+i\varepsilon)\right]^{-2}$$

$$- \left[x\left((p-k)^2-\Lambda^2+i\varepsilon\right)+(1-x)(k^2-m^2+i\varepsilon)\right]^{-2}$$

Let's simplify some of the expressions:

$$\gamma_\mu(\not{k}+m)\gamma^\mu = -2\not{k} + 4m$$

$$x\left[(p-k)^2-\mu^2+i\varepsilon\right]+(1-x)(k^2-m^2+i\varepsilon) = xp^2 + xk^2 - 2xp\cdot k - \mu^2 x + xi\varepsilon + k^2 - xk^2 - (1-x)m^2 + i\varepsilon - xi\varepsilon$$

$$= i\varepsilon + k^2 - 2xp\cdot k + x^2p^2 - x^2p^2 + xp^2 - \mu^2 x - m^2(1-x) = (k-xp)^2 + x(1-x)p^2 - (1-x)m^2 - \mu^2 x + i\varepsilon$$

let's define $(k-xp) = l$ and $\Delta \equiv -x(1-x)p^2 + (1-x)m^2 + \mu^2 x$

$$\Rightarrow \quad x\left[(p-k)^2-\mu^2+i\varepsilon\right]+(1-x)(k^2-m^2+i\varepsilon) = l^2 - \Delta + i\varepsilon$$

the shift of $l = k - xp$ means $dk = dl$

Now we simplify: $x\left[(p-k)^2-\Lambda^2+i\varepsilon\right]+(1-x)(k^2-m^2+i\varepsilon) = xp^2+xk^2-2xp\cdot k-\Lambda^2 x+i\varepsilon x+k^2-xk^2-m^2(1-x)+i\varepsilon-i\varepsilon x$

$$= i\varepsilon + k^2 - 2xp\cdot k + x^2p^2 - x^2p^2 + xp^2 - \Lambda^2 x - m^2(1-x) = l^2 - \Delta_\Lambda + i\varepsilon$$

where $\Delta_\Lambda = -x(1-x)p^2 + (1-x)m^2 + \Lambda^2 x$ (also $\not{k} = \not{l} + x\not{p}$)

$$\Rightarrow \quad -i\Sigma_2(p) = -e^2 \int \frac{d^4l}{(2\pi)^4} \int_0^1 dx \left(-2(\not{l}+x\not{p})+4m\right)\left[\frac{1}{(l^2-\Delta+i\varepsilon)^2} - \frac{1}{(l^2-\Delta_\Lambda+i\varepsilon)^2}\right]$$

The term in numerator proportional to $\not{l}$ integrates to zero so we can set it zero.

$$\Rightarrow \quad -i\Sigma_2(p) = -e^2 \int_0^1 dx \int \frac{d^4l}{(2\pi)^4} \frac{-2x\not{p}+4m}{(l^2-\Delta+i\varepsilon)^2} - \frac{-2x\not{p}+4m}{(l^2-\Delta_\Lambda+i\varepsilon)^2}$$

Now we Wick rotate via $l^0 \to i l^0_E$ and $l^i = l^i_E \Rightarrow dl^0 \Rightarrow i\,dl^0_E$

$$\Rightarrow \quad \underset{\text{Minkowski}}{\frac{d^4l}{(2\pi)^4}} \to i \underset{\text{Euclid}}{\frac{d^4l_E}{(2\pi)^4}} \qquad \| \; l^2 = l^{0^2} - l^{i^2} \to -l_E^2$$

$$\Rightarrow \quad -i\Sigma_2(p) = -e^2 \int_0^1 dx \; i \int \frac{d^4l_E}{(2\pi)^4} \frac{-2x\not{p}+4m}{(-l_E^2-\Delta+i\varepsilon)^2} - \frac{-2x\not{p}+4m}{(-l_E^2-\Delta_\Lambda+i\varepsilon)^2}$$

To evaluate the Euclidean integral we no-longer need $i\varepsilon$-prescription so we can set it to 0

$$\Rightarrow \quad \Sigma_2(p) = 2e^2 \int_0^1 dx \; (-x\not{p}+2m) \int \frac{d^4l_E}{(2\pi)^4} \frac{1}{(l_E^2+\Delta)^2} - \frac{1}{(l_E^2+\Delta_\Lambda)^2}$$

We can now use the spherical coordinates $\int d^4l_E = \int d\Omega_4 \int_0^\infty d|l_E|\,|l_E|^3$

with $\int d\Omega_4 = 2\pi^2$ and $\int_0^\infty dl_E \, l_E^3 \frac{1}{(l_E^2+\Delta)^2} = \frac{1}{2}\left(\frac{\Delta}{\Delta+l_E^2} + \ln(l_E^2+\Delta)\right)\Big|_{l_E=0}^{l_E=\infty}$

We have done this integral in previous exercise so using the result directly

For the term $\frac{1}{(l_E^2-\Delta_\Lambda)^2}$ we get same answer but with Δ_Λ instead of Δ

$$\Rightarrow \quad \Sigma_2(p) = \frac{4\pi^2 e^2}{(2\pi)^4}\int_0^1 dx \, (-x\not{p}+2m) \, \frac{1}{2}\left[\frac{\Delta}{\Delta + l_E^2}\Bigg|_{l_E=0}^{l_E=\infty} - \frac{\Delta_\Lambda}{\Delta_\Lambda + l_E^2}\Bigg|_{l_E=0}^{l_E=\infty} + \ln\left[\frac{l_E^2+\Delta}{l_E^2+\Delta_\Lambda}\right]\Bigg|_{l_E=0}^{l_E=\infty}\right]$$

$$\lim_{l_E\to\infty} \ln\left(\frac{l_E^2+\Delta}{l_E^2+\Delta_\Lambda}\right) = \ln(1) = 0 \quad , \quad \lim_{l_E\to\infty} \frac{\Delta}{\Delta + l_E^2} = \frac{\Delta}{\infty} = 0$$

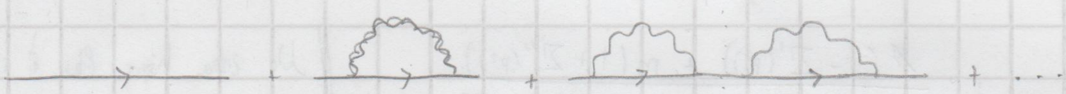
$$\Rightarrow \quad \Sigma_2(p) = \frac{1}{8\pi^2} e^2 \int_0^1 dx \, (-x\not{p}+2m)\left(-\ln\left[\frac{\Delta}{\Delta_\Lambda}\right]\right) \qquad \Bigg\| \quad \frac{e^2}{4\pi} = \alpha$$

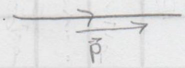
$$\Sigma_2(p) = \frac{\alpha}{2\pi}\int_0^1 dx \, (-x\not{p}+2m)\ln\left[\frac{\Delta_\Lambda}{\Delta}\right]$$

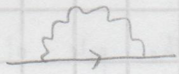
$\Delta_\Lambda = x\Lambda^2 + (1-x)m^2 - x(1-x)p^2$ so in limit of $\Lambda \to \infty$ the leading contribution is $x\Lambda^2$. We can keep only this contribution when looking at UV divergence sine $\Lambda \to \infty$ as a result we get:

$$\Sigma_2(p) = \frac{\alpha}{2\pi}\int_0^1 dx \, (-x\not{p}+2m)\ln\left[\frac{x\Lambda^2}{-x(1-x)p^2 + (1-x)m^2 + x\mu^2}\right] \quad \square$$

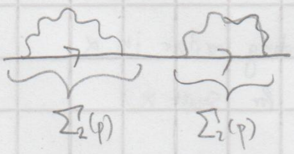
Exercise 6.3



The first diagram is  $= \dfrac{i(\not{p}+m)}{p^2-m^2}$

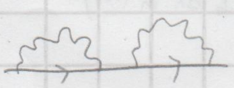
Second diagram is  $= \dfrac{i(\not{p}+m)}{p^2-m^2}\left[-i\Sigma_2(p)\dfrac{i(\not{p}+m)}{p^2-m^2}\right]$

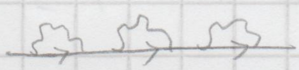
We can factor the third diagram as a "multiplication" of two Second diagrams



$$= \frac{i(\not{p}+m)}{p^2-m^2}\left[-i\Sigma_2(p)\right]\frac{i(\not{p}+m)}{p^2-m^2}\left[-i\Sigma_2(p)\right]\frac{i(\not{p}+m)}{p^2-m^2}$$

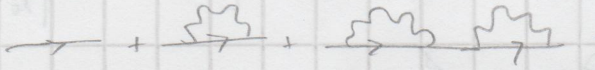
We can use $i\not{p}\,\Sigma_2(p) = i\Sigma_2(p)\,\not{p}$ since $\not{p}^2 = p^2$ and only Gamma-matrix term in $\Sigma_2(p)$ is $\not{p}$ so $[\Sigma_2(p), \not{p}] = 0$

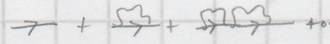
$\Rightarrow$  $= \dfrac{i(\not{p}+m)}{p^2-m^2}\left[-i\Sigma_2(p)\dfrac{i(\not{p}+m)}{p^2-m^2}\right]^2$

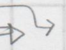
And if we have  then we just get one more $i\Sigma_2(p)\dfrac{i(\not{p}+m)}{p^2-m^2}$ term times previous 3-loop diagram so this progresses as geometric series.
A geometric series can be written as $a + aq + aq^2 + aq^3 + aq^4 + \cdots = a\sum_n^\infty q^n$

$$\Rightarrow \quad a\sum_{n=0}^{\infty} q^n = \frac{a}{1-q}$$ for $|q|<1$, we assume $|q|<1$ since we are describing

a physical process.

$\Rightarrow$  $+\cdots = \dfrac{i(\not p + m)}{p^2 - m^2}\left[1 - \left(-i\cdot i \Sigma_2(p)\,\dfrac{(\not p + m)}{p^2 - m^2}\right)\right]^{-1}$

$$\frac{1}{\not p - m} = \frac{(\not p + m)}{p^2 - m^2} \Rightarrow$$  $+\cdots = \dfrac{i}{\not p - m}\cdot\dfrac{1}{1 - \dfrac{\Sigma_2(p)}{\not p - m}}$

$\Rightarrow$  $= \dfrac{i}{\not p - m - \Sigma_2(p)}$

From given definitions for $\Sigma'(p^2)$ and $\Sigma''(p^2)$ and the form we calculated for $\Sigma_2(p)$ we see that $\Sigma_2(p) = \not p\,\Sigma'(p^2) + m\,\Sigma''(p^2)$

$$\Rightarrow \frac{i}{\not p - m - \Sigma_2(p)} = \frac{i}{\not p - m - \not p\,\Sigma'(p^2) - m\Sigma''(p^2)} = \frac{i}{\not p(1-\Sigma'(p^2)) - m(1+\Sigma''(p^2))}$$

$$= i\,\frac{\not p(1-\Sigma'(p^2)) + m(1+\Sigma''(p^2))}{\not p^2(1-\Sigma'(p^2))^2 - m^2(1+\Sigma''(p^2))^2} \qquad \Big\| \quad \not p^2 = p^2$$

$$\frac{i}{\not p - m - \Sigma_2(p)} = i\,\frac{\not p(1-\Sigma'(p^2)) + m(1+\Sigma''(p^2))}{p^2(1-\Sigma'(p^2))^2 - m^2(1+\Sigma''(p^2))^2}$$

No idea how the i is supposed to disappear.

Exercise 6.4

The physical mass $p^2 = m_e^2$ defined as the pole of the Propagator:

The pole is: $\left. p^2(1-\Sigma'(p^2))^2\right|_{p^2=m_e^2} - m^2(1+\Sigma''(p^2))^2 = 0$

$\Sigma' \sim \alpha$ and $\Sigma'' \sim \alpha$ so we can look at this to the leading order in α since α is small:

taylor: $(1+x)^\alpha \approx 1 + \alpha x$ for small x

$$m_e^2 = m^2(1+\Sigma''(p^2))^2(1-\Sigma'(p^2))^{-2} \approx m^2\left[1 + 2\Sigma''(p^2) + \mathcal{O}(\alpha^2)\right]\left[1 + 2\Sigma'(p^2) + \mathcal{O}(\alpha^2)\right]$$

$$= m^2\left[1 + 2\Sigma''(p^2) + 2\Sigma'(p^2) + \mathcal{O}(\alpha^2)\right]$$

Now insert $\Sigma' = -\frac{\alpha}{2\pi}\int_0^1 dx\, x \ln\{\cdots\}$

$\Sigma'' = \frac{\alpha}{\pi}\int_0^1 dx \ln\{\cdots\}$

$$m_e^2 = m^2\left[1 + \frac{\alpha}{2\pi}\int_0^1 dx\,[4-2x]\ln\left[\frac{x\Lambda^2}{-x(1-x)p^2 + (1-x)m^2 + x\mu^2}\right]\right]$$

To the leading order in α we can write $p^2 \approx m^2$ inside the integral

$$\Rightarrow -x(1-x)p^2 + (1-x)m^2 \approx (1-x)^2 m^2$$

$$\Rightarrow \quad m_e^2 = m^2\left[1 + \frac{\alpha}{2\pi}\int_0^1 dx\,[4-2x]\ln\left(\frac{x\Lambda^2}{(1-x)^2 m^2 + x\mu^2}\right)\right] \blacksquare$$

Near the pole the propagator has the form

$$Z_2 \frac{i(\not p + m)}{p^2 - m^2}$$

Since $\Sigma \sim \alpha$ is small we get:

$$\frac{i}{\not p - m - \Sigma_2(p)} \approx \frac{i}{(\not p - m)\left(1 - \frac{\partial \Sigma_2}{\partial \not p}\big|_{m_e}\right)} = \frac{i(\not p + m)}{p^2 - m^2}\frac{1}{1 - \frac{\partial\Sigma_2(p)}{\partial \not p}\big|_{m_e}}$$

$$\Rightarrow Z_2 = \left(1 - \frac{\partial \Sigma_2(p)}{\partial \not p}\Big|_{m_e}\right)^{-1} \approx 1 + \frac{\partial \Sigma_2(p)}{\partial \not p}\Big|_{m_e} = 1 + \frac{\partial \Sigma_2(p)}{\partial \not p}\Big|_m$$ since

we can approximate $m \approx m_e$ inside the dx integral to the first order in α.

$$\frac{\partial \Sigma_2(p)}{\partial \not p} = \frac{\partial}{\partial \not p}\left[\frac{\alpha}{2\pi}\int_0^1 dx\,(-x\not p + 2m)\ln\left[\frac{x\Lambda^2}{-x(1-x)p^2 + (1-x)m^2 + x\mu^2}\right]\right]$$

$$p^2 = \not p \not p = \not p^2 \qquad \text{and } \frac{d}{dx}\ln\left[\frac{a}{bx+j}\right] = \frac{d}{dx}\left[\ln a - \ln[bx+j]\right] = -\frac{\frac{d}{dx}(bx)}{bx+j}$$

$$\Rightarrow \frac{\partial}{\partial \not p}\Sigma_2(p)\Big|_m = \frac{\alpha}{2\pi}\left[-\int_0^1 dx\; x\ln\left[\frac{x\Lambda^2}{-x(1-x)p^2 + (1-x)m^2 + x\mu^2}\right] - (-x\not p + 2m)\frac{2(-x(1-x))\not p}{-x(1-x)p^2 + (1-x)m^2 + x\mu^2}\right]\Bigg|_m$$

Boundary condition $\not p = m \Rightarrow (-x\not p + 2m)\not p\,\big|_m = (2-x)m^2$

$$\Rightarrow Z = 1 + \frac{\alpha}{2\pi}\int_0^1 dx\left[-x\ln\left(\frac{x\Lambda^2}{(1-x)^2 m^2 + x\mu^2}\right) + (2-x)\frac{2m^2 x(1-x)}{(1-x)^2 m^2 + x\mu^2}\right]$$

$$\Rightarrow Z = 1 + \frac{\alpha}{2\pi}\int_0^1 dx\left[-x\ln\left(\frac{x\Lambda^2}{(1-x)^2 + x\mu^2}\right) + (2-x)\frac{2m^2 x(1-x)}{(1-x)^2 m^2 + x\mu^2}\right] \blacksquare$$

The first integral in expression of Z is UV-divergent so we can approximate the logarithm as $\ln\left(\frac{\Lambda^2}{m^2}\right)$, which gives us

$$\frac{\alpha}{2\pi}\int_0^1 dx\left(-x\ln\left(\frac{\Lambda^2}{m^2}\right)\right) = -\frac{1}{2}\frac{\alpha}{2\pi}(1^2 - 0^2)\ln\frac{\Lambda^2}{m^2} = -\frac{1}{2}\frac{\alpha}{2\pi}\ln\left[\frac{\Lambda^2}{m^2}\right]$$

The second integral is $\quad IR = \frac{\alpha}{2\pi}2m^2\int_0^1 dx\,(1 + (1-x))\frac{(1-(1-x))(1-x)}{(1-x)^2 m^2 + x\mu^2}$

$$IR = \frac{\alpha}{\pi}m^2\int_0^1 dx\,\frac{[1-(1-x)^2](1-x)}{(1-x^2)m^2 + x\mu^2} = \frac{\alpha}{\pi}\int_0^1 dx\,\frac{1-x}{(1-x^2) + x\frac{\mu^2}{m^2}}\left[1 + O(1-x)\right]$$

Let's evaluate this to the first order in $(1-x)$, since $1-x$ is small as $x \to 1$, and at $x=0$ our integral is just zero because integrand is $(2-x)\frac{2m^2 x(1-x)}{(1-x)^2 m^2 + x\mu^2}$

$$\Rightarrow IR \simeq \frac{\alpha}{\pi}\int_0^1 dx \, \frac{1-x}{(1-x)^2 + x\frac{\mu^2}{m^2}}$$

$$(1-x)^2 + x\frac{\mu^2}{m^2} = 1-2x+x^2+x\frac{\mu^2}{m^2} = x^2 + x\left(\frac{\mu^2}{m^2}-2\right)+1$$

This has root at

$$x_0 = \frac{-\left(\frac{\mu^2}{m^2}-2\right) \pm \sqrt{\left(\frac{\mu^2}{m^2}-2\right)^2-4}}{2}$$

$$x_{01} = -\left(\frac{\mu^2}{2m^2}-1\right) - \frac{1}{2}\sqrt{\left(\frac{\mu^2}{m^2}-2\right)^2-4}, \quad x_{02} = -\left(\frac{\mu^2}{2m^2}-1\right) + \frac{1}{2}\sqrt{\left(\frac{\mu^2}{m^2}-2\right)-4}$$

$$\Rightarrow (1-x)^2 + x\frac{\mu^2}{m^2} = \left(x+\left[\frac{\mu^2}{2m^2}-1\right] - \frac{1}{2}\sqrt{\left(\frac{\mu^2}{m^2}-2\right)^2-4}\right)\left(x+\left[\frac{\mu^2}{2m^2}-1\right] + \frac{1}{2}\sqrt{\left(\frac{\mu^2}{m^2}-2\right)^2-4}\right)$$

Let's do change of variables: $\left(\frac{\mu^2}{m^2}-2\right) \equiv b$ and $x + \frac{1}{2}b = u \Rightarrow x = u - \frac{1}{2}b$

$\Rightarrow du = dx$ and u has integration limits from $0+\frac{1}{2}b$ to $1+\frac{1}{2}b$

$$\Rightarrow IR = \frac{\alpha}{\pi}\int_{\frac{1}{2}b}^{1+\frac{1}{2}b} du \, \frac{1-u+\frac{1}{2}b}{\left(u-\frac{1}{2}\sqrt{b^2-4}\right)\left(u+\frac{1}{2}\sqrt{b^2-4}\right)} \quad \Big\| \quad k \equiv \frac{1}{2}\sqrt{b^2-4}$$

$$= -\frac{\alpha}{\pi}\int_{\frac{1}{2}b}^{1+\frac{1}{2}b} du \, \frac{1-u+\frac{1}{2}b}{(k-u)(k+u)} \quad \Big\| \quad \frac{1}{(k-u)(u+k)} = +\frac{1}{2k}\left[\frac{1}{k-u}+\frac{1}{k+u}\right]$$

$$= -\frac{\alpha}{2\pi k}\int_{\frac{1}{2}b}^{1+\frac{1}{2}b} du \, \left(1+\frac{1}{2}b-u\right)\left[\frac{1}{k+u}+\frac{1}{k-u}\right]$$

$$= -\frac{\alpha}{2\pi k}\left(1+\frac{1}{2}b\right)\Big[\ln[k+u]-\ln[k-u]\Big]\Bigg|_{\frac{1}{2}b}^{1+\frac{1}{2}b} + \frac{\alpha}{2\pi k}\int_{\frac{1}{2}b}^{1+\frac{1}{2}b} du \, u\left[\frac{1}{k+u}+\frac{1}{k-u}\right]$$

$$\ln(k+u) = \ln\left(k\left(1+\frac{u}{k}\right)\right) = \ln(k) + \ln\left(1+\frac{u}{k}\right) \quad \text{and}$$

$$\ln(k-u) = \ln\left(k\left(1-\frac{u}{k}\right)\right) = \ln(k) + \ln\left(1-\frac{u}{k}\right)$$

$$\Rightarrow \ln(k+u) - \ln(k-u) = \ln\left(1+\frac{u}{k}\right) - \ln\left(1-\frac{u}{k}\right)$$

and

$$\int du \, \frac{u}{k+u} + \frac{u}{k-u} = -k\ln(k+u) + u - k\ln(k-u) - u = -k\ln(k^2-u^2)$$

$$\Rightarrow IR = -\frac{\alpha}{2\pi}\left(1+\frac{1}{2}b\right)\left[\ln\left(1+\frac{1+\frac{1}{2}b}{k}\right) - \ln\left(1+\frac{\frac{1}{2}b}{k}\right) - \ln\left[1-\frac{1+\frac{1}{2}b}{k}\right] + \ln\left[1-\frac{\frac{1}{2}b}{k}\right]\right]$$

$$- \frac{\alpha}{2\pi}\left[\ln\left(\frac{k^2-\left(1+\frac{1}{2}b\right)^2}{k^2-\left(\frac{1}{2}b\right)^2}\right)\right]$$

First let's ~~investigate the term~~ what happens at infrared (IR)-divergence: $\mu \to 0$, let's start by investigating term $\frac{1+\frac{1}{2}b}{k}$

$$\frac{1+\frac{1}{2}b}{k} = \frac{1+\frac{1}{2}b}{\frac{1}{2}\sqrt{b^2-4}} = \sqrt{\frac{(2+b)^2}{(b+2)(2-b)}} = \sqrt{\frac{2+b}{2-b}}$$

IR-divergence $\mu \to 0 \Rightarrow b \to -2 \Rightarrow \lim_{b\to -2}\sqrt{\frac{2+b}{2-b}} = \sqrt{\frac{0}{4}} = 0$

While this term does multiply 4-logarithms we must first see what happens to the logarithms as $b \to -2$ since if they diverge we need to investigate further:

$$\ln\Big(1+\underbrace{\frac{1+\frac{1}{2}b}{k}}_{=0 \text{ as } b\to 0}\Big) = \ln(1) = 0 \qquad \text{simirarily: } \ln\Big(1-\frac{1+\frac{1}{2}b}{k}\Big) = 0$$

$$\ln\Big(1+\frac{\frac{1}{2}b}{k}\Big) = \ln\Big(1-k+\underbrace{\frac{1+\frac{1}{2}b}{k}}_{=0}\Big) = \ln(1-k) \quad \Big\Vert \text{ and } k = \tfrac{1}{2}\sqrt{b^2-4} = 0 \text{ as } b \to 0$$

$$\ln\Big(1+\frac{\frac{1}{2}b}{k}\Big) = \ln(1) = 0 \quad \text{and} \quad \ln\Big(1-\frac{\frac{1}{2}b}{k}\Big) = \ln\Big(1+k-\frac{1+\frac{1}{2}b}{k}\Big) = \ln(1) = 0$$

So all the logarithms that multiply $\frac{\alpha}{2\pi}\cdot\frac{1+\frac{1}{2}b}{k}$ are 0 so there is no bad behaviour on the first part of the expression IR

now let's investigate the logaritm

$$\ln\Big[\frac{k^2-(1+\frac{1}{2}b)^2}{k^2-(\frac{1}{2}b)^2}\Big] = \ln\Big[\frac{k^2-(1+\frac{2}{2}b+\frac{1}{4}b^2)}{k^2-b^2/4}\Big] \quad \Big\Vert\; k^2 = \tfrac{1}{4}(b^2-4) = \tfrac{b^2}{4}-1$$

$$\Rightarrow \; = \ln\Big[\frac{-2-b}{-1}\Big] = \ln[b+2] \quad \Big\Vert\; b = \frac{\mu^2}{m^2}-2$$

$$= \ln\Big[\frac{\mu^2}{m^2}\Big]$$, we have found the IR-Divergence of the Z_2

$$\Rightarrow Z_2 = 1 - \frac{1}{2}\frac{\alpha}{2\pi}\ln\Big[\frac{\Lambda^2}{m^2}\Big] + IR = 1 - \frac{1}{2}\frac{\alpha}{2\pi}\ln\Big[\frac{\Lambda^2}{m^2}\Big] - \frac{\alpha}{2\pi}\ln\Big[\frac{\mu^2}{m^2}\Big]$$

$$Z_2 = 1 + \frac{\alpha}{2\pi}\Big[-\frac{1}{2}\ln\frac{\Lambda^2}{m^2} + \ln\frac{m^2}{\mu^2}\Big]$$

According to LSZ-theorem each external leg comes with $\sqrt{Z_2}$

$\Rightarrow$ Cross section get $Z_2^2 \approx 1 + \frac{\alpha}{2\pi}\big(-\ln\frac{\Lambda^2}{m^2} + 2\ln\frac{m^2}{\mu^2}\big)$ + Finite to $\mathcal{O}(\alpha)$

(we ignore terms of order α^2)

So the external leg cross section becomes $d\sigma^{ext} = d\sigma^\circ Z_2^2 = d\sigma^\circ \frac{\alpha}{2\pi}\Big[-\ln\frac{\Lambda^2}{m^2} + 2\ln\frac{m^2}{\mu^2}\Big]$ + finite terms

We also have $d\sigma^{rad} + d\sigma^{vertex} = \text{Finite} + d\sigma^\circ\cdot\frac{\alpha}{2\pi}\Big(\ln\big(\frac{\Lambda^2}{m^2}\big) - 2\ln\big(\frac{m^2}{\mu^2}\big)\Big)$ (from last week)

so we get

$$d\sigma^{ext} + d\sigma^{rad} + d\sigma^{vertex} = \text{finite} + d\sigma^\circ\cdot\frac{\alpha}{2\pi}\Big(\underline{-\ln\big(\tfrac{\Lambda^2}{m^2}\big)} + 2\ln\big(\tfrac{m^2}{\mu^2}\big)\Big) + d\sigma^\circ\frac{\alpha}{2\pi}\Big(\underline{\ln\big(\tfrac{\Lambda^2}{m^2}\big)} - 2\ln\big(\tfrac{m^2}{\mu^2}\big)\Big)$$

$\Rightarrow$ The infinite terms cancel each other.

$\Rightarrow \quad d\sigma^{rad} + d\sigma^{vert} + d\sigma^{ext} = \text{finite number} \quad \square$

$\Rightarrow$ We have shown that the UV and IR divergence from the expression of Z_2 leads to finite expression for cross-section.